TV Lift Firmware Operational Information V1.0 071024

Default State:

- **Attractor Video**: The unit will display an attractor video (vidattr) on a loop via the transparent TV.
- **Position**: TV is in the default (bottom) position.
- **Input Wait**: Awaiting customer interaction via the capacitive touch panel button.

Customer Interaction:

- **Button Press**:
  - **Video Change**: Display vid1.
  - **Movement**: TV begins moving to the top position.
  - **LED Lights**: White LED lights on the H frame flash.
  - **Synchronization**: Monitor actuator travel to ensure synchronization.

Top Position Actions:

- **Media Playback**: Play vid2 upon reaching the top position.
- **Customer Presence**:
  - **Check Presence**: After vid2 ends, check the presence sensor to detect if a customer is in front of the screen.

If Customer is Present:

- **Loop Video**: Continue looping vid2.
- **Periodic Check**: Every 10 seconds, verify customer presence.
- **Extended Presence**: If the customer remains for 3 minutes, initiate TV movement to the bottom position.

If Customer is Not Present:

- **Play Vid3**: Start playing vid3.
- **Beam Break Sensor**:
  - **Object Detection**: Check the beam break sensor for objects in the TV's path.
  - **Object Detected**: Display a warning image instructing removal of objects from the surface under the TV.
  - **Safety Check**: Every 5 seconds, recheck the beam break sensor to ensure it is safe to travel.

Safe Travel:

- **Movement**: Initiate TV movement to the bottom position.
- **LED Lights**: White LED lights on the H frame flash.

- **Synchronization**: Monitor actuator travel to ensure synchronization.
- **Continuous Monitoring**: Continuously monitor the beam break sensor for obstructions.
- **Completion**:
    - **Successful Descent**: Once the TV reaches the bottom position, switch media player back to vidattr.
    - **Reset State**: Unit returns to waiting for customer input.